Summative essay: Can AI systems be intelligent?

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Ethics of Data and AI PH440

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Feedback reflection: This summative essay responded to and changed every point made in the formative feedback. The concept of intelligence was clearly described as having different dimensions rather than as a general overarching term. The equivalence made between the human brain and AI algorithms was changed to an analogy to better fit this picture of intelligence. The two arguments put forward for the analogy approach were made distinct in the conclusion.

I Introduction

Artificial Intelligence undoubtedly has the ability to solve a multitude of "complex and cognitively demanding problems" (Chemero 2023, p.1828). AI-driven taxis successfully navigate city streets, bringing customers from one place to another (BBC 2024, p.1). AI systems can accurately predict protein structures, significantly contributing to scientific progress. This breakthrough solves a long-unsolved scientific problem that traditional methods have failed to address (Jumper et al. 2021, p.2). ChatGPT can pass the prestigious BAR law exam (Rose 2025, p.1). Do these "seemingly intelligent behaviours" (Mitchell and Krakauer 2023, p.1) mean that AI is intelligent?

In this essay, I argue that artificial intelligence is intelligent in a fundamentally different way from humans, endorsing the view that intelligence comes in different types. First, I introduce the internal workings of AI using large language models. Then, I present two distinct reasons for AI cognition being different to human: 1. AI's lack of conceptual meaning and 2. Limits of Datafication. Finally, I argue that these differences in cognition underpin, what is, a different type of intelligence.

II Us vs Them

AI is fundamentally based on prediction. For example, a large language model (LLM), a type of AI, is an algorithm which generates text responses to human inputs. It achieves this by generating statistically probable word sequences by passing vast training data through a neural network (Chemero 2023, p.1828). This can be seen in the training of the ChatGPT 3 model, which used 45 terabytes of pure text data (OpenAI, Brown, et al. 2020, p.8). If an LLM is given the sentence "Mandela's first name is", its autoregressive nature will predict probabilities for the next word based on a function of only the previous words in the sentence. In this case, one would expect the word given the highest probability to be "Nelson" and therefore put it in the sentence. It is in this way that AI algorithms work. Instead of text, a Robotaxi is trained only on images and sensor data collected from human drivers to make a decision on the road, and biological imaging and data are used to predict protein structures.

It is relatively commonplace for analogies to be drawn between the human brain and algorithms. Human brains rely on biological mechanisms (like neurons and synapses) while algorithms use computational ones. Each learns from data: humans through sensory input and life experiences, LLMs through large-scale text. Learning in both involves feedback- brains adapt via neural plasticity; LLMs adjust through back propagation (Hahari 2015, p.124).

This analogous picture, combined with AI's incredible ability to hold a conversation and provide information, has some in the AI research community arguing that they truly understand the language and can perform human-like reasoning (Mitchell and Krakauer 2023, p.2). In the next two sections, I provide 2 reasons why this analogous approach to understanding AI intelligence is fundamentally flawed.

III In Search of True Meaning

Scenario: John Searle imagines himself in a locked room receiving Chinese symbols from the outside. He does not understand Chinese, but he follows English instructions to respond to the outside with

the correct Chinese symbols. To outsiders, it seems like he understands, but in reality, he has no real understanding of the Chinese language; he is just following written rules (Searle 1980, p.418-423).

Neural language models, in their current form, are bound to statistical models. The sentences created are purely formed through the prediction of the next word based on training data. This is best illustrated by OpenAI's neural scaling power laws, which describe a predictable increasing but diminishing relationship between model performance and factors including model size, computing power, data and parameters (see figure 1).

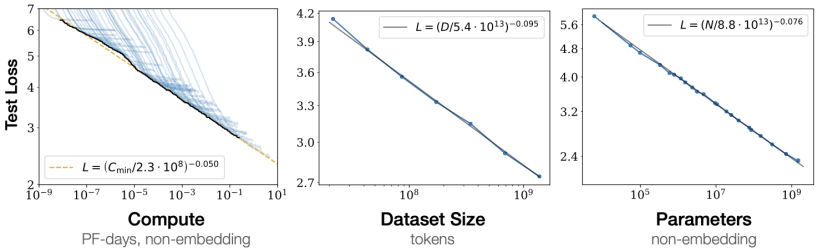

Figure 1: OpenAI's scaling laws (OpenAI, Kaplan, et al. 2020, p.3)

These factors establish a hard *quantitative* limit on current AI performance, in the same way as the English instructions provide a hard *quantitative* limit on Searle's ability to respond to outside Chinese symbols. I.e., A longer list of instructions would give Searle more nuance and accuracy for responding in Chinese to the outside.

In comparison, the human brain seems to have an "elusive ingredient" (Hahari 2015, p.153). "Human-like understanding" does not seem to be based on the massive statistical models AI systems are based on. One leading theory for how the brain operates is that it is based on concepts; mental models of "external categories, situations, and events of one's own internal state and self" (Mitchell and Krakauer 2023, p.3). These mental models provide humans with "functional" language abilities, allowing us to understand the language and relate it to the real world; make analogies, generalisations and reason compositionally or counterfactually, going beyond statistical properties of linguistics (Mitchell and Krakauer 2023, p.3).

From the outside, the outputs generated by Searle in his room are identical to those created by someone with a true understanding of the language, and the same can be said when applied to a single one-shot AI model. On the other hand, the power laws, derived from application in many models, show that, despite huge quantitative augmentation in performance, current models don't go beyond the brute-force pattern learning, which creates these quantitative shifts. A true semantic understanding of meaning would entail a *qualitative* understanding of language, which could be represented by a model which produces results below the power law lines.

Therefore, the popular approach to current AI development, which is embodied by the mantra: "scale is all you need" (Mitchell and Krakauer 2023, p.2) is misplaced if the goal is conceptual meaning. The seemingly human-like feats achieved by AI, which many view as being a result of human-like understanding, can be credited to "shortcut learning". This is where a system relies on correlations in data, rather than human-like understanding, to perform well on a particular benchmark. Interestingly, there has been some literature on the possible nature of meaning and concepts being derived from neural dynamics (Tononi, Gerald, and Sporns 1998; Tononi 2004). Furthermore, there is still the hope, in the industry, that sufficiently large models will give rise to

conceptual structures. But in its current form, AI fundamentally differs from human intelligence in its lack of conceptual understanding to the tasks it is undertaking.

IV The Limits of Datafication

The understanding of human cognition being concept-based is a prominent theory, yet it is by no means the only valid perspective, with many scientists taking the analogy between an algorithm and the human brain even further. The "narrating self" view suggests that decisions are made unconsciously through biological and neural processes, in the same way an algorithm does, with the self only constructing concepts and narratives after the fact, for post-hoc explanation (Hahari 2015, p.351).

Yet, even taking this demystified understanding of human cognition, the limits of datafication will always create a demarcation between human and AI understanding. Datafication is the "ability to render into data many aspects of the world that have never been quantified before" (Cukier and Mayer-Schoenberger 2013, p.29). This data is then fed into the training and utilisation of an algorithm to infer probabilities.

Humans are embodied, which is to say that "we are living, biological creatures who are in constant interaction with the material social, cultural and technological environment" (Chemero 2023, p.1829). The physical features and activities of our bodies do not merely influence cognition; they constitute it. Our metabolic needs and the perpetual micro-movements of our eyes, neck, and torso, all directed toward acting and perceiving the world in a perception-action loop, are integral parts of the system that is cognitive processing. For example, when we catch a ball, we do not compute its parabola symbolically; our visual system (input) and arm co-regulate in real-time. Chemero points out that current AI's are not fully embodied in the way humans are (Chemero 2023, p.1828). But, there has been an increase in embodiment in the artificial intelligence industry. In 2024, NVIDIA introduced a foundational model pitched as an open model for generalised reasoning and skills, with a reusable "motor+ perception" function (NVIDIA 2024, p.1). Google has also introduced sensorimotor inputs into its 540-billion-parameter LLM (Google 2023, p.1). These AI systems go beyond disembodied text prediction and are moving toward living and acting in the world they are a part of.

As AI gets better at tasks, it will likely continue to be further embodied into its material, social and technological environment. However, even in the highly likely scenario of increased AI embodiment, I argue that AI will never be embodied in the way that humans are embodied due to the limitations of datafication. Because every signal an AI receives must be datafied, parsed into countable symbols - pixels, voltage readings, joint-angle values- it lives inside a world which has been mathematised. The vague unease that accompanies a stranger's stare, the warmth of shared laughter, the feeling of listening to your favourite song; certain embodied human pleasures and pains are pragmatically or fundamentally irreducibly qualitative, depending on one's view of human cognition, and cannot be fed into an algorithm for decision-making, $n \neq$ all for human inputs (to borrow from Cukier and Mayer-Schoenberger 2013, p.29). It is in this asymmetry in data inputs that the boundary lies between machine embodiment and human embodiment, and therefore between human cognition and AI cognition.

V Intelligence: Us and them

These two distinct differences between human cognition and AI cognition, the argument from true understanding and from datafication, are to show that "our way of being intelligent is not the only way and not their way" (Chemero 2023, p.1829).

- AI systems, by using correlations, can solve problems which humans apply conceptual understanding to solve.
- AI systems will understand the nature of any problem very differently from the data it can datafy for its embodiment, i.e., perception through eyes and ears will be different from perception through a camera and microphone.

Intelligence is widely defined as the "ability to solve problems", as an *outcome* (Chemero 2023, p.1828; Hahari 2018, p.85). AI systems are undoubtedly intelligent; I could continue this essay indefinitely with examples of AI solving problems. However, it is important to frame intelligence not as a unitary property, but as a family of different types to understand AI's ability to solve problems. AI's datafied and correlation-based intelligence could be considered as a "novel form of understanding", with its "alien" intuition and ability to pass human benchmarks (Mitchell and Krakauer 2023, p.3-4). It would simply be a "category error" to associate our human-embodied, concept-based intelligence with the intelligence underpinning AI, especially as it continues to show impressive performance without it. This expanding menagerie of cognitive architectures—from octopus nervous system to transformer stacks—invites a pluralistic understanding of intelligence, where the difference is descriptive, not deficient. This framing of intelligence doesn't smuggle in the assumption that problem-solving prowess must always be underwritten by the same internal cognition.

The newest shifts in AI benchmarks shift focus away from human linguistic imitation to respond to this plurality in intelligence. For example, MMLU-pro demands chain-of-thought reasoning (Wang et al. 2024, p.2), ARC-AGI demands causal reasoning on abstract grid puzzles (ARC 2025, p.1). This is in response to possible emergent conceptual structures from "scale is all you need" or neural dynamics approaches.

VI Conclusion

In conclusion, there are two distinct differences between human and AI intelligence. 1. The lack of true understanding is AI's current inability to go beyond quantitative statistics, with the data it has been given. Although there is some hope being placed in large-scale models, or neural dynamics, AI currently does not have this conceptual understanding. 2. The data an AI can use to reason (statistically, conceptually, or something else) will be fundamentally different from that of a human. Even when embodied in its surroundings, it will be embodied in a different way to humans. These distinctions highlight the differences in types of intelligence, allowing us to better understand how different solutions are found to different problems.

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